NIKHIL MANKALI



AI Engineer | AWS Certified Cloud Practitioner

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https://d1t6fdfo6c7b31.cloudfront.net Siegen, DE (Will Relocate)

SUMMARY

AI Engineer experienced in building and deploying production-ready AI and LLM-based applications, with a focus on delivering business value through intelligent systems. Skilled in developing APIs, integrating machine learning models into applications, and deploying scalable solutions on AWS and Azure. Experienced in building AI-driven features such as chatbots, decision-support tools, and automated workflows. Strong focus on translating business needs into reliable, maintainable, and user-facing AI systems.

SKILLS

Programming

Python, C++, SQL, Bash, HTML, CSS, JavaScript

ML/DL Frameworks

PyTorch, TensorFlow, HuggingFace, OpenCV, NumPy, Pandas, Matplotlib, Scikit-learn, Seaborn, Tensorboard, Weights & Biases, MLFlow

Computer Vision

Object Detection, Segmentation, Anomaly Detection, OOD Detection, Adversarial Robustness, Uncertainty Estimation, Aerial Perception, Image Processing

LLM/GenAI

NLP, LLMs, Prompt Engineering, Transformers, LLM Fine-tuning, Model Serving, OpenAPI, Claude Code, Copilot, Whisper, Ollama, OpenRouter, OpenCode

RAG & Agentic AI Frameworks

LangChain, LangGraph, Amazon Bedrock Agentcore, FastMCP, FastAPI, Streamlit

Databases & Retrieval Systems

PostgreSQL, MongoDB, FAISS, Neo4j, ChromaDB, Azure AI Search

Cloud & Deployment

AWS, Azure, Docker, Kubernetes, Grafana, Prometheus, Terraform (IaC), CloudFormation(IaC), GitHub, GitHub Actions, GitLab

EXPERIENCE

05/2025 - Present

Siegen, Germany

AI Engineer - AgenticAI, RAG Systems, LLMOps

Independent Projects, Proof of Concepts

- **Travel Agent** - Conversational AI trip planner that gathers requirements and orchestrates parallel web-search workers to produce a real-time itinerary.
- **MCP Servers -** Built production grade MCP servers using FastMCP for local and remote deployment. For Example: Expense Management MCP Server, Audio Transcribing MCP server using Whisper
- **RAG Based Portfolio Website** - Built a portfolio website with a serverless RAG Chatbot using AWS (Lambda, Bedrock, S3, API Gateway, CloudFront), with React frontend.
- **Smart Receipt Scanner -** Built a serverless receipt scanner web application using AWS (Lambda, S3, DynamoDB, API Gateway, Cognito) with React frontend, featuring OCR text extraction, multi-user authentication, and budget analytics with expense categorization.
- **RAG Chatbot -** Developed a production-grade retrieval-augmented Q&A system using Azure OpenAI, Azure AI Search, and FastAPI; achieved 0.89 faithfulness (RAGAS) and improved answer relevance by 23% through A/B testing. Deployed as a containerized microservice on Azure Container Apps with automated CI/CD pipelines.
- **Web Scraper -** Created a full-stack scraper to curate research papers from Springer API, generating multilingual summarized PDF newsletters via ChatGPT API and Python scripting. Delivered secure authentication, translation, and PDF generation with a table of contents to improve accessibility and research workflow.
- **Tech Stack:** Python, PyTorch, LangChain, LangGraph, Strands, FastAPI, FastMCP, Docker, AWS, Azure, React, HTML, CSS.

11/2023 - 04/2025

Ingolstadt, Germany

Research Assistant - Computer Vision, Anamoly Detection

Fraunhofer Institute for Transport and Infrastructure Systems IVI

- Designed and deployed an AI-based anomaly detection system for aerial perception, enabling reliable decision-making in real-world applications.
- Validated the module in real-world drone flight tests; successfully flagged unseen objects (e.g., dogs) absent in training datasets.
- Integrated uncertainty estimation into the CV pipeline to improve model performance, robustness, and reliability in real-world scenarios.
- Built and maintained end-to-end AI pipelines (data processing, model training, evaluation, and deployment) using Python, Docker, and Kubernetes, ensuring production-ready and scalable solutions.
- Contributed ML models and tools that were part of internal project pitches and funding proposals for UAM research.
- **Tech Stack -** Python, PyTorch, HuggingFace, Docker, Kubernetes, Grafana, Bash, Git, CI/CD.

EXPERIENCE

06/2023 - 09/2023
Ingolstadt, Germany

Research Assistant - Data Science
Technische Hochschule Ingolstadt

- Collected and cleaned datasets with various view types (top-down, frontal, oblique) to create a unified aerial vision dataset.
- Balanced dataset coverage and ensured annotation quality for training object detection and segmentation models.
- Resulted in improved accuracy and generalization of downstream computer vision tasks at Fraunhofer IVI.
- **Tech Stack -** Python, PyTorch, Bash, Git, CI/CD.

08/2022 - 06/2023
Siegen, Germany

Research Assistant - Computer Vision
Computer Vision Department - Universität Siegen

- Investigated adversarial robustness of CV models by simulating attacks on grayscale, RGB, and 3D voxel data.
- Improved robustness using adversarial training: FGSM accuracy increased to 95%, PGD to 46%.
- **Tech Stack -** Python, PyTorch, SLURM, Git, Bash, Distributed Computing.

INTERNSHIP

06/2023 - 09/2023
Ingolstadt, Germany

Computer Vision Intern
Fraunhofer Institute for Transport and Infrastructure Systems IVI

- Analyzed and harmonized 15+ aerial datasets used for CV model training, focusing on annotation format, class distribution, and input normalization.
- Wrote scripts to automate label conversion and enhance compatibility with object detection and segmentation frameworks.
- Created internal documentation on Docker image creation, Kubernetes deployment via kubectl, and Octane simulation usage, now used as onboarding material.
- Collaborated with cross-functional teams to integrate AI modules into existing workflows, supporting testing, deployment, and real-world application scenarios.
- **Tech Stack -** Python, PyTorch, Docker, Kubernetes, Git, Bash, Object Detection, Segmentation.

EDUCATION

07/2025 - 10/2025
Hamburg, Germany

AWS re/Start Cloud Computing Bootcamp
neuefische GmbH - School and Pool for Digital Talent

- Networking, Linux Systems, Terraform, Scripting, Docker, Kubernetes, AWS Services

10/2020 - 03/2024
Siegen, Germany

M.Sc. Mechatronics
Universität Siegen

GPA **1.8** / **4**

- **Subjects:** Software Engineering (C++, Python, SQL), Unsupervised Deep Learning, Computer Vision, Artificial Intelligence, Automation and Industrial Communication, Project Management
- **Master's Thesis:** Pixel-wise Out-of-Distribution Detection in Semantic Segmentation for Aerial Imagery using Normalising Flow Networks for Negative Feature Synthesis
- **Studienarbeit:** Evaluating and improving the robustness of deep neural networks through adversarial training

06/2015 - 05/2019
Hyderabad, India

B.Tech. Mechanical Engineering
CVR College of Engineering

GPA **9.63** / **10**

- **Subjects:** Engineering Mathematics, Fluid Mechanics, Mechanics, IC Engines, CAD, CNC, Finite Element Methods.

CERTIFICATIONS

AWS Certified Cloud Practitioner (CLF-C02)
AWS [Link]

LANGUAGES

German
Advanced (B1/B2)



English
Fluent (C1/C2)

